Schematic Symbols & Conventions

The alphabet of electronics before you touch a CAD tool

Resistors • Capacitors • ICs • Power rails • Nets

Audience: Beginner Engineering Students

IEEE / ANSI / IEC styles • Reference designators • Hierarchical sheets

Why Standardised Symbols Matter

Every engineer should read the same diagram the same way

Universal language

An Indian, German, or Japanese engineer reads the same drawing.

Auditability

Compliance reviewers, fab houses, legal — all rely on consistency.

CAD interchange

EDA tools swap netlists if everyone follows IEEE/IEC conventions.

Speed

Familiar symbols → faster reading → fewer review errors.

Education

Textbooks, datasheets, app-notes all use standard graphical conventions.

History

Schematic conventions go back over 100 years — proven, refined.

Two Major Style Families

IEEE/ANSI (American) vs IEC (European)

IEEE / ANSI — Zigzag style

- Resistor = zigzag line
- Capacitor = two short bars
- Inductor = bumps / coils
- Common in US textbooks, MCU vendors
- Default in KiCad's older libraries

IEC 60617 — Rectangle style

- Resistor = rectangle
- Capacitor = two short bars (same)
- Inductor = filled half-circles or rect.
- Common in Europe, India, Asia
- Default in many EDA tools' newer libraries

Reference Designators

The names you give every part — IEEE 315 conventions

Prefix	Component	Examples
R	Resistor	R1, R2, R47, R_PULLUP
C	Capacitor	C1, C2 … C100
L	Inductor (Coil)	L1, L_FB1
D	Diode (incl. LEDs, Zeners, TVS)	D1, D_TVS1
Q	Transistor (BJT, FET)	Q1, Q2
U	Integrated Circuit	U1, U2 … U_MCU
J	Jack / Connector	J1, J_USB
P	Plug (mating connector)	P1, P_PWR
Y	Crystal / Oscillator	Y1
X	Transducer (general)	X1
S / SW	Switch	SW1, S_PWR
F	Fuse	F1
FB	Ferrite Bead	FB1
TP	Test Point	TP1, TP_GND
RV	Variable Resistor (Pot)	RV1, RV_VOL

Wires, Nets, Junctions & No-Connects

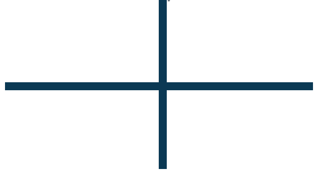
How connectivity is drawn

Connected (junction)

A dot means the wires join.

Crossing (no connection)

No dot → wires pass over.



Net label

Wires of same name = same net.

No-Connect (NC)

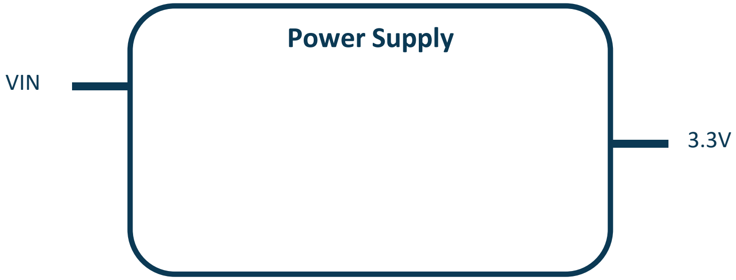
Intentionally unconnected.

Bus & bus entries

Thick line carries multiple nets. Use bus entries to pull individual nets in/out.

Hierarchical sheet

A 'block' in a parent sheet whose internals live in another file.

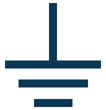


Power & Ground Symbols

Many shapes — same meaning

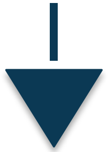
Power and ground symbols connect by NET NAME, not graphics — choose one style per project

Earth / Chassis



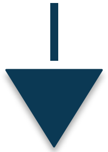
Connects to chassis or mains earth

Signal Ground



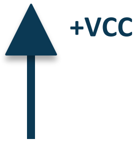
Common 0 V reference for analog signals

Digital Ground



Often split from analog GND

VCC / VDD



Positive supply rail

VEE / VSS

Negative supply (op-amps)

Multiple rails

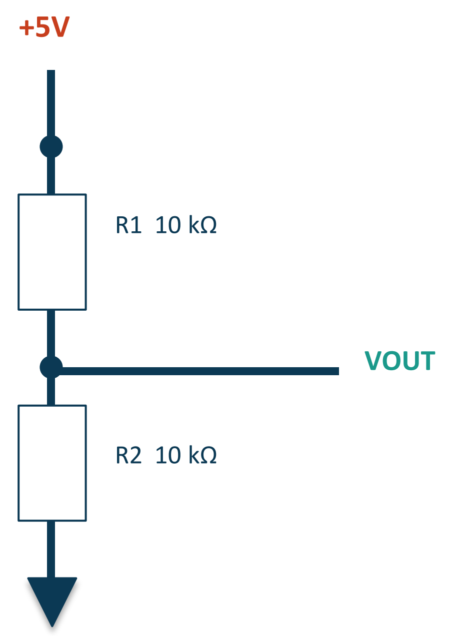
+3.3V / +5V / +12V — labelled

Annotation & Drawing Conventions

Habits that turn a schematic into a readable document

- Inputs on the LEFT, outputs on the RIGHT
- Power rails go UP, ground goes DOWN
- Component values next to the symbol — not on the wire
- Reference designators above or to the left of the symbol
- Pin 1 marker visible on every multi-pin IC
- Use net labels for any wire crossing the sheet boundary
- Group related parts (e.g. each LDO with its caps) together
- Title block on every sheet — designer, date, rev, file name

What a 'tidy' schematic looks like



Inputs left, outputs labelled, power up, ground down — readable in seconds.

Hierarchical Schematics

Splitting a complex design into pages

When to split

- Schematic > 50 parts — split it
- By function: Power / MCU / Sensors / I/O
- Each sub-sheet has clear ports (in/out nets)
- Parent sheet shows the block diagram
- Reusable blocks: copy across projects

Naming & organisation

- One .kicad_sch file per page
- Sheet names: power.sch, mcu.sch, io.sch
- Ports = hierarchical labels (in / out / passive)
- Global labels for board-wide nets (VCC, GND)
- Title block + revision on every page

Cross-References, Tables & Tables of Variants

Pro touches that elevate a schematic

Off-page connectors

Mark where a net continues on another sheet — net name + sheet ref.

Bus expansion

D[0..15] → D0, D1, … D15 via bus entries.

Variant tables

Cell A: 50 Ω, Cell B: 75 Ω — for stuff-options.

Test points

Add TP_GND, TP_VCC, TP_DEBUG to every design.

Footnotes

Use footnotes for non-obvious choices (e.g. 0 Ω = jumper).

Title block

Designer / Reviewer / Date / Rev / Filename — non-negotiable.

Common Schematic Mistakes

What design reviews flag in the first 5 minutes

Floating nets

Net label with no other endpoint = ERC error.

Missing power flags

ERC needs to see VCC / GND defined.

Wrong symbol

BJT used where MOSFET intended — looks similar.

No pin 1 marker

Layout flips the IC silently.

Inconsistent names

+3V3, 3.3V, +3.3V all mean the same — DRC sees them differently.

Skipped ref-des

R1, R2, R4 — looks unprofessional, breaks BOM tools.

Schematic Review Checklist

What to look for in 30 minutes

- Block diagram visible at top of parent sheet
- Title block complete on every page
- All power rails clearly labelled & flagged
- Every IC has decoupling caps on its power pins
- Reset / boot / debug pins visible
- Pull-ups / pull-downs on every floating input
- Crystal load caps present & sized
- ESD / TVS on every external connector

- All connectors have ferrite + cap filters
- Test points on all critical nets
- Fuses / current limiters on power inputs
- Indicator LEDs (power, status) present
- BOM auto-export tested
- ERC: zero errors, zero unintentional warnings
- Reviewer signed off on title block
- Schematic file is in version control

Schematic Capture Tools

What the industry uses today

Tool	Cost	Pros	Cons
KiCad	Free / OSS	Full PCB suite, Python scripting	Steeper learning curve
Altium Designer	₹3L+/year	Industry standard, polished UI	Expensive seat license
EasyEDA	Free / cloud	Built-in JLCPCB ordering	Cloud lock-in
Eagle / Fusion	Subscription	Tight Fusion 360 integration	Owned by Autodesk now
OrCAD / Allegro	₹5L+	Enterprise tier, mixed-signal sim	Heavy for small projects
Mentor Xpedition	Enterprise	High-end, FPGA-team friendly	Overkill for most
DipTrace	₹15–60k	Mid-tier, good for SMEs	Smaller community

Recommendation for students

Start with KiCad. It's free, professional-grade, runs on every OS, and what you learn there transfers directly to Altium when you join industry.

Reading Other People's Schematics

A learnable skill — and a huge career multiplier

- Skim the block diagram first — understand functional sections
- Find power rails — what voltages, where they come from
- Find the MCU/main IC — what does it talk to, what controls it
- Trace from connector → ESD/filter → buffer → MCU
- Read decoupling caps last — they're rarely interesting but always present
- Resist the urge to scan pin-by-pin. Work block-by-block
- Open-source projects (Adafruit, Arduino, Olimex, RPi) are excellent practice
- Bonus: try recreating a published schematic in KiCad to test understanding

Best Practices Recap

12 habits that separate juniors from seniors

- One style (IEEE OR IEC) per project — never mix
- Inputs on left, outputs on right
- Power up, ground down
- Hierarchical sheets for > 50 parts
- Net names use UPPER_CASE with underscores
- Decoupling caps near every IC power pin
- Add test points & fiducials
- Use no-connect markers explicitly
- Run ERC every save
- Title block on every sheet
- Version-control schematics (Git-friendly text format)
- Get a peer to review — every revision

Key Takeaways

Key takeaways from this lecture

Symbols = language

Learn the alphabet — read industry schematics worldwide.

Connectivity discipline

Net labels, junctions, and ref-des are non-negotiable.

Hierarchy scales

Split big designs into functional pages.

Review every save

ERC + peer review catch 80% of bugs free.

Tools follow conventions

KiCad / Altium just enforce IEEE/IEC — learn the rules.

Practice on real designs

Read open-source schematics weekly.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- IEEE Standard 315 — Graphical Symbols for Electrical Diagrams
- IEC 60617 — international counterpart
- kicad.org / docs.kicad.org — official KiCad tutorials
- Adafruit / SparkFun / Olimex — open-source schematics to read
- RPi.org — Raspberry Pi schematic PDFs are excellent reading practice
- Robert Feranec (YouTube) — schematic & PCB walkthroughs
- EEVblog (YouTube) — schematic-reading lessons
- Altium / KiCad official docs — naming conventions, hierarchical design

Thank you • Build something this week.